

SMART INDIA HACKATHON 2026

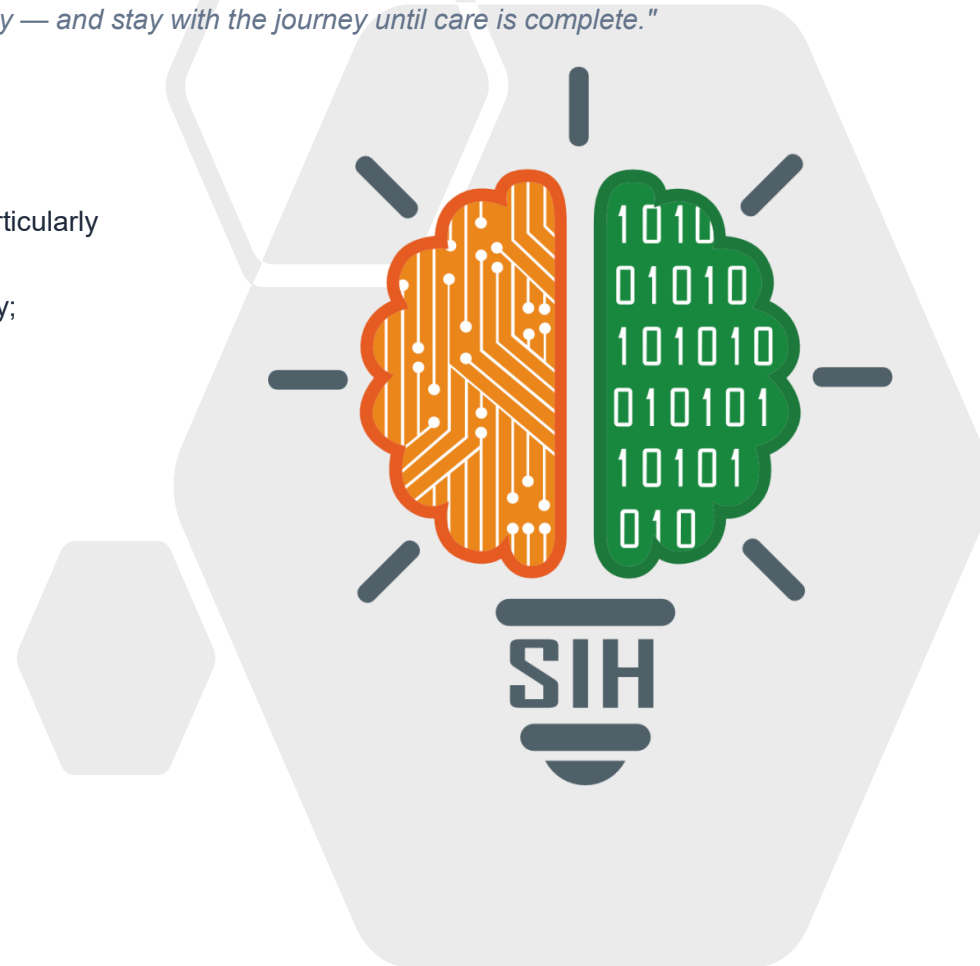
SWASTHYASETU (स्वास्थ्यसेतु)



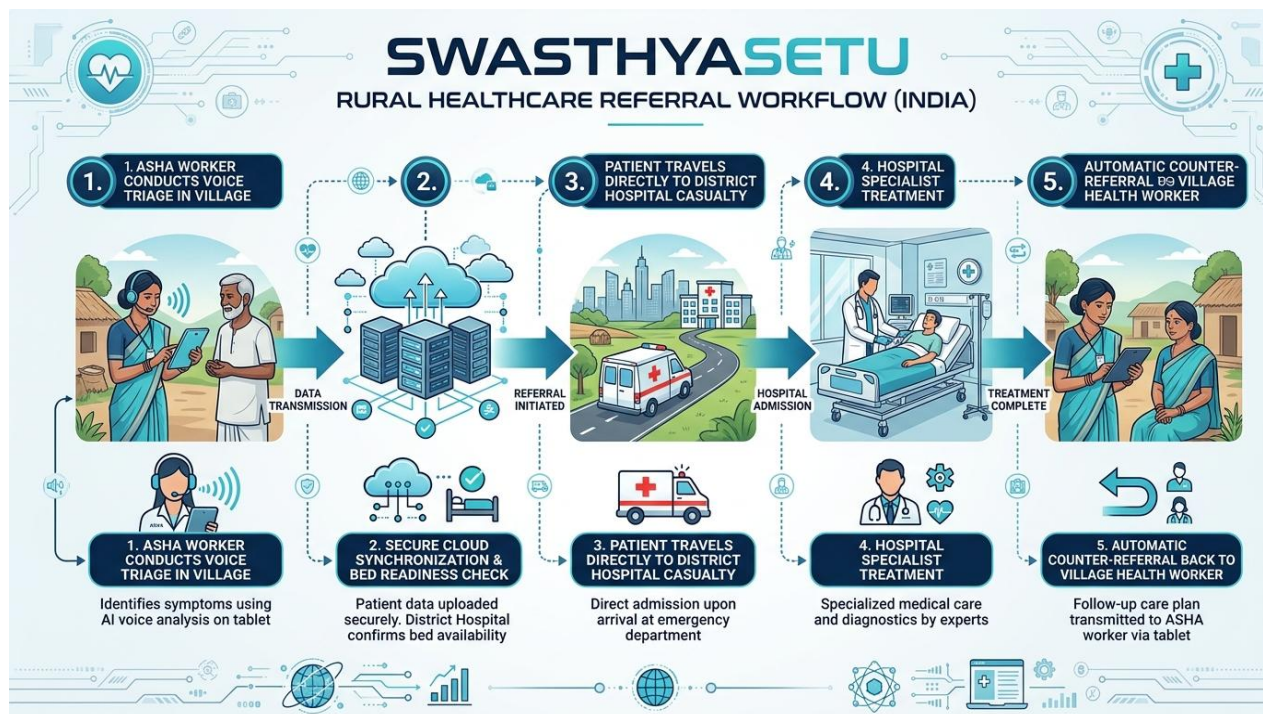
Rural Healthcare Referral & Care Continuity Platform

"Don't just send the patient. Make sure the care is ready — and stay with the journey until care is complete."

- **Problem Statement ID:** SIH26133
- **Problem Statement Title:** Accessibility and quality of public healthcare services, particularly in rural and underserved areas
- **Nodal Authority:** Government of Maharashtra (Maharashtra State Innovation Society; Department of Skills, Employment, Entrepreneurship and Innovation)
- **Theme:** MedTech / BioTech / HealthTech
- **PS Category:** Software
- **Team ID:** SIH2026-TEAM-2
- **Team Name:** SwasthyaSetu (Team 2)



Proposed Solution (Describe your Idea/Solution/Prototype) — Detailed Explanation | Problem Resolution | Uniqueness & Innovation

**KEY NOVELTY: The ONLY Closed-Loop Public Health Referral Platform**

- Bidirectional Care Visibility: Village Sub-Centre (AAM) → PHC → CHC → District Hospital → ASHA Counter-Referral.
- National Digital Health Rails: Native ABDM UHI (FHIR R4), ABHA creation, and auto-triage for MJPJAY cashless coverage.

THE PROBLEM: The Rural 'Clinical Black Hole'

- 150,000+ blind paper referrals weekly across India's 31,000 PHCs without confirmation.
- 48.2% drop-out: Patients travel 40-80 km only to find specialists absent or beds full.
- Zero Counter-Referral: Discharges never reach village ASHAs, causing fatal relapses.

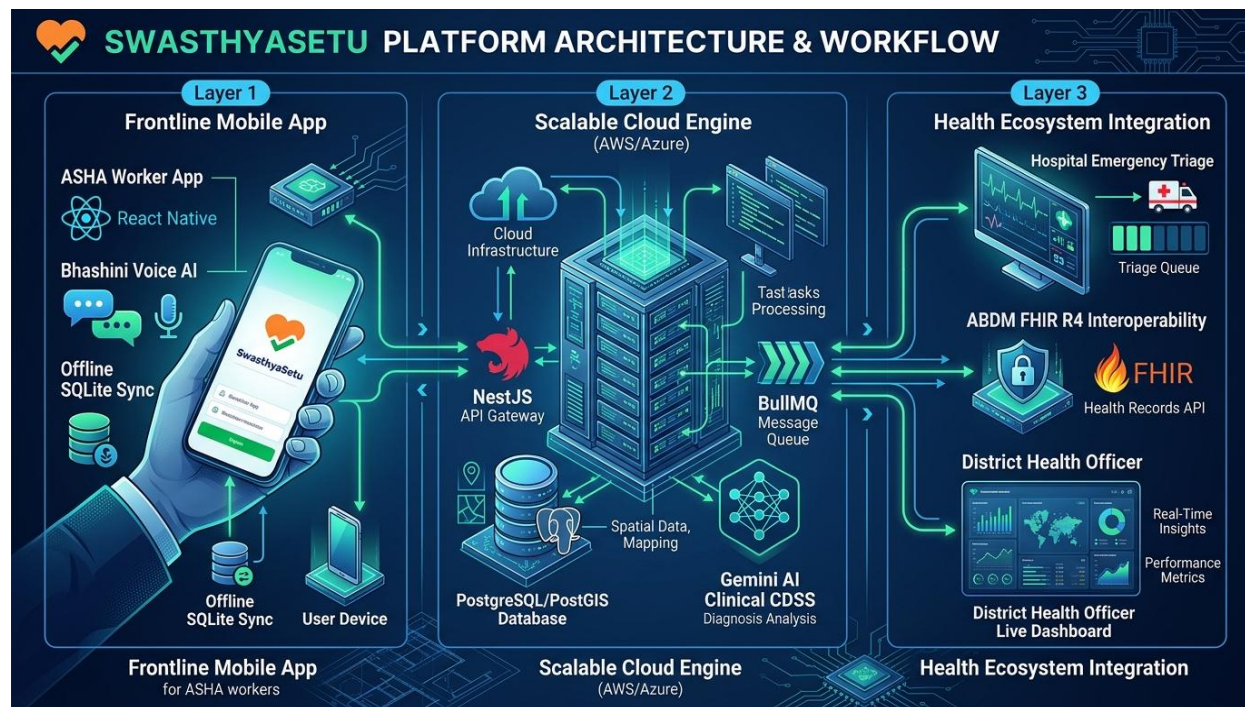
THE SOLUTION: AI-Powered Closed-Loop Care Orchestration

- End-to-End Orchestration: Real-time patient tracking from village clinic to tertiary discharge.
- Care Readiness Verification: 6-factor check ensures specialist, ICU bed, & diagnostics ready.
- Zero-Data-Loss: Multilingual Marathi voice intake + offline-first encrypted sync.

4 ARCHITECTURAL PILLARS (Frontline to Tertiary)

- **Pillar 1 (Voice-First)**: Bhashini Marathi/Hindi speech-to-text; 50s intake on 2GB RAM phones.
- **Pillar 2 (Offline-First)**: SQLite/WatermelonDB CRDTs store 5,000+ records in monsoon blindspots.
- **Pillar 3 (One-Trip Care)**: Bundles specialist OPD, ECG/CBC labs & transit into a single visit.
- **Pillar 4 (Readiness WOW)**: Algorithmic pre-departure verification of doctor presence & ICU beds (<85%).

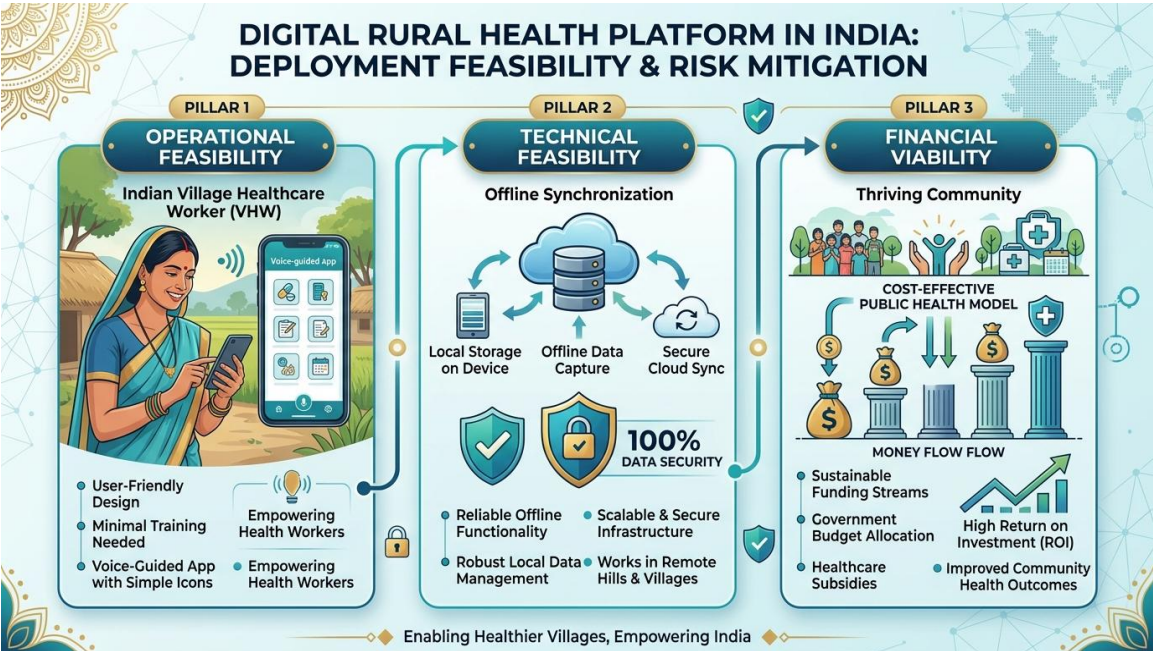
Technologies Used (Languages, Frameworks, Architecture) & Methodology Process (Flowcharts / Working Prototype Pipeline)

**6-STAGE CARE ORCHESTRATION PIPELINE (WORKING PROTOTYPE)**

- 01. Intake (Voice): ASHA speaks Marathi; Bhashini STT + OCR slip ingests vitals offline.
- 02. CDSS & Bundle: ICMR/WHO CDSS flags triage severity (Red/Amber) & bundles care.
- 03. Readiness Check: Algorithmic 6-factor check (Doctor, Beds, Diagnostics) at destination.
- 04. Dynamic Routing: Spatial PostGIS Haversine matrix routes patient; sends DLT SMS token.
- 05. Fast-Track OPD: Hospital pre-notified; patient scans QR token, bypassing 3.5h queues.
- 06. Counter-Referral: Tertiary discharge auto-creates home follow-up task on village ASHA app.

Layer	Production Tech Stack	Engineering Rationale
Mobile & Web	Vite + React 18 PWA / React Native Android Go	Ultra-lightweight (<15 MB APK), offline-ready, runs on 2GB RAM phones.
API Gateway	NestJS (Node.js 20+) Modular Microservices	High-throughput async REST/GraphQL, JWT + Argon2 security, DLT SMS.
Persistence & GIS	PostgreSQL 16 + PostGIS + Prisma ORM	ACID transactions, spatial hospital distance queries, drive-time radius.
Offline & Queue	SQLite / WatermelonDB CRDTs + Redis BullMQ	Guaranteed local storage during network drops; background sync workers.
AI & Speech	Gemini 2.5 Flash + Bhashini Indic STT + OCR	Sub-second symptom extraction, vernacular Marathi voice dictation.
Protocols & Security	ABDM Gateway (ABHA, HFR), FHIR R4, DPDP Act	National Health Authority compliant, encrypted health bundles, patient consent.

Analysis of Idea Feasibility, Potential Challenges & Risks, and Strategic Mitigation Measures



SUSTAINABLE B2G FINANCIAL MODEL & OPERATIONAL VIABILITY

- Operational Feasibility: Replaces 8 duplicate paper registers; works on existing smartphones.
- Sustainable Funding: Funded via State NHM PIP Innovation budget + ABDM DHIS incentives (₹500/bed/mo).
- District Unit Economics: ~₹18 Lakh/year district operating cost vs ~₹35 Lakh/year state contract.
- 48.5% operating margin directly funds village frontline worker training and field engineering support.

Risk Category	Identified Challenge	Strategic Mitigation Plan
Technical Risk	Cellular blackout >7 days in deep tribal monsoon areas.	Local SQLite retains 5,000 records encrypted; offline QR token + peer-to-peer sync at weekly PHC meet.
Operational Risk	Hospital specialist ignores incoming referral queue.	Auto-escalation alert to Civil Surgeon dashboard after 4 hours + SMS reminder + physical priority token.
Clinical Risk	Triage rule misclassifies acute case as low risk.	Deterministic ICMR/WHO CDSS thresholds; mandatory human-in-the-loop doctor authorization; no autonomous diagnosis.
Adoption Risk	Low smartphone literacy among senior village ASHAs.	Marathi voice dictation + visual iconographic UI + ₹50 performance incentive per verified referral follow-up.

Potential Impact on Target Audience & Comprehensive Benefits (Social, Economic, Environmental, Administrative)

< 10%

Referral Drop-out

Slashed from 48.2% baseline via closed-loop tracking.

60%+

OOPE Travel Cut

Saves families from debt traps via Coordinated One-Trip.

< 45 Min

Hospital Wait Time

Fast-track casualty triage bypasses 3.5h general queues.

100%

Follow-up Compliance

Automated counter-referral tasks sent to village ASHAs.

TRANSFORMING RURAL HEALTHCARE IN
INDIA BY **SWASTHAYSETU**

FOUR KEY IMPACT PILLARS

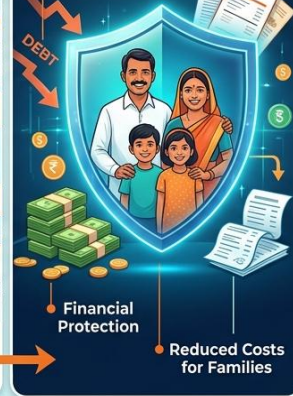
1. LIVES SAVED

- Maternal and Child Health
- Timely Interventions



2. OUT-OF-POCKET
EXPENSE REDUCTION

- Financial Protection



- Financial Protection
- Reduced Costs for Families

3. FAST HOSPITAL
TRIAGE

- Efficient Patient Flow
- Quick Care Delivery



- Efficient Patient Flow
- Quick Care Delivery

4. DATA GOVERNANCE

- Real-Time Hospital Bed Availability
- Advanced Analytics



- Real-Time Hospital Bed Availability
- Advanced Analytics

SOCIAL IMPACT: Maternal & Cardiac Lives Saved

Prevents maternal deaths (PPH, eclampsia) & cardiac crises in tribal blocks (Melghat, Nandurbar); vernacular voice guides illiterate families; empowers ASHAs.

ECONOMIC IMPACT: Eliminating Healthcare Debt

Cuts private transport and redundant diagnostic fees that push 55M into poverty yearly; unlocks ABDM DHIS hospital cash incentives (₹500/bed/mo).

ENVIRONMENTAL IMPACT: Paperless Green Health

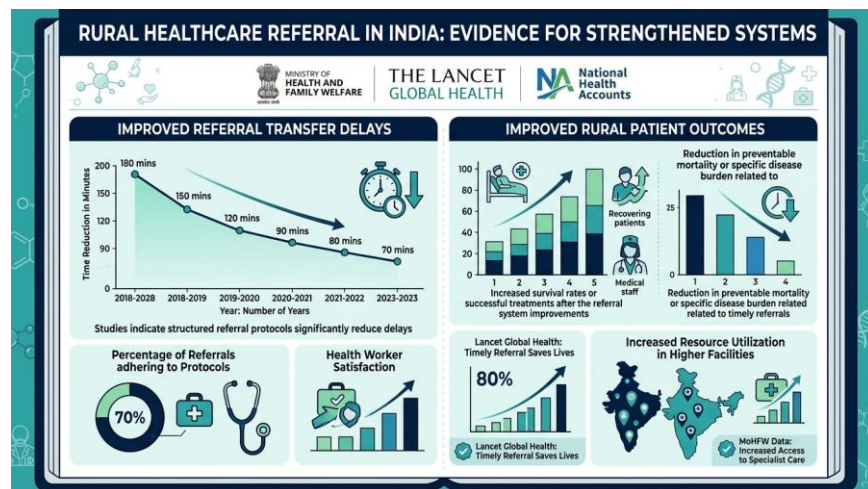
Replaces millions of paper registers with digital health records; slashes transit vehicular emissions from repeated, failed hospital trips.

ADMINISTRATIVE IMPACT: Real-Time Governance

Live District Health Officer (DHO) bottleneck heatmaps surface specialist shortages & equipment outages; replaces delayed 45-day HMIS reports.

Details & Links of Reference Research Work, Competitive Benchmarking Matrix, and Epidemiological Evidence

Key Capability Dimension	eSanjeevani (C-DAC)	NIC e-Hospital	Khushi Baby / CHT	108 EMS Fleet	SwasthyaSetu
Closed-Loop Physical Referral	✗ None (Tele-advice only)	✗ None (Internal OPD only)	△ Partial (MCH only)	✗ Transport only	✓ Full Closed-Loop Tracking
Offline-First Mobile Architecture	✗ Fails without 4G	✗ Desktop Intranet only	✓ Offline Sync	△ Voice Call based	✓ Offline SQLite + CRDT Sync
Vernacular Voice Triage	✗ None	✗ None	✗ None	✗ None	✓ Bhashini Marathi/Hindi STT
Pre-Arrival Casualty Alert	✗ None	✗ None	✗ None	△ Manual Phone Call	✓ Real-Time Triage Card
Automated ASHA Counter-Referral	✗ None	✗ None	✗ None	✗ None	✓ Discharge to ASHA Task



1. MoHFW Rural Health Statistics

Source: Health Dynamics of India 2022-23

Finding: Acute 79.9% shortfall of specialists at rural CHCs.

Relevance: Validates why capacity-aware routing is essential.

2. The Lancet Global Health

Source: Referral Pathways in Rural India (2020)

Finding: 48.2% of referred primary patients never reach hospital.

Relevance: Direct empirical baseline for closed-loop tracking.

3. National Health Accounts

Source: NHA 2020-21 / Lancet Public Health

Finding: OOPE is 47.1% of health spend; pushes 55M into debt.

Relevance: Solved by Coordinated One-Trip Care Bundles.

4. JMIR 2025 & ABDM Specs

Source: Li et al., JMIR 2025; ABDM FHIR R4

Finding: Bidirectional referral cuts transfer delay to 0.90 days.

Relevance: Blueprint for SwasthyaSetu ABHA integration.